

US Patent & Trademark Office

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United States Patent Application Publication

20250279509

Kind Code

A1

Publication Date

September 04, 2025

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MANUFACTURING METHOD FOR ELECTRICAL ENERGY STORAGE DEVICE

Abstract

A manufacturing method for an electrical energy storage device disclosed herein includes: a preparing step of preparing a plate material that is made of metal and includes a first part forming a first surface of a case, and a pair of second parts extending from the first part and forming a pair of second surfaces of the case; and a wrapping step of disposing an electrode body on the first part of the plate material and bending the plate material so that a distance between the pair of second parts decreases, thereby wrapping the electrode body.

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Family ID: 96860878

Appl. No.: 19/061928

Filed: February 24, 2025

Foreign Application Priority Data

JP 2024-031736

Mar. 01, 2024

Publication Classification

Int. Cl.: H01M50/103 (20210101); B21D5/16 (20060101); B23K26/20 (20140101); H01G11/84 (20130101); H01M10/0525 (20100101); H01M10/0585 (20100101); H01M10/0587 (20100101); H01M50/119 (20210101)

U.S. Cl.:

CPC H01M50/103 (20210101); B21D5/16 (20130101); B23K26/20 (20130101); H01G11/84 (20130101); H01M10/0525 (20130101); H01M10/0585 (20130101); H01M10/0587

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to Japanese Patent Application No. 2024-031736 filed on Mar. 1, 2024. The entire contents of this application are hereby incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

1. Field

[0002] The present disclosure relates to a manufacturing method for an electrical energy storage device.

2. Background

[0003] In general, an electrical energy storage device includes an electrode body and a case that accommodates the electrode body. One of the references of the conventional art related to a manufacturing method for an electrical energy storage device is Japanese Patent No. 4537353. Japanese Patent No. 4537353 describes to manufacture a square battery by preparing a case main body whose cross sectional shape is like a letter U with one surface open, inserting an electrode body into the case main body through the opening, and then closing the opening with a sealing plate.

SUMMARY

[0004] According to the present inventor's knowledge, the electrode body is often formed to be as large as possible within the range of being able to be inserted through the opening of the case main body from the viewpoints of improving the volume energy density, and the like. Therefore, in some cases, when the electrode body is inserted through the opening, an outer peripheral part of the electrode body is caught at the opening and the accommodation has been difficult. In addition, at the insertion, the electrode body may be in contact with the opening, which results in a crease in a separator of the electrode body or displacement of an electrode plate.

[0005] The present disclosure has been made in view of the above circumstances, and an object is to provide a novel manufacturing method for an electrical energy storage device in which an accommodation property of an electrode body is improved.

[0006] The present disclosure provides a manufacturing method for an electrical energy storage device including one or a plurality of electrode bodies and a case that has a square shape and accommodates the electrode body or the electrode bodies, in which the case includes at least a first surface, a pair of second surfaces extending from a pair of edges of the first surface and facing each other, and a third surface facing the first surface. This manufacturing method includes: a preparing step of preparing a plate material that is made of metal and includes at least a first part forming the first surface of the case, and a pair of second parts extending from the first part and forming the pair of second surfaces of the case; and a wrapping step of disposing the electrode body on the first part of the plate material and bending the plate material so that a distance between the pair of second parts decreases, thereby wrapping the electrode body.

[0007] In the present disclosure, the plate material is bent so as to wrap the electrode body in the wrapping step; therefore, the accommodation property of the electrode body can be improved relatively compared to a conventional method in which an electrode body is inserted through an opening on one surface of a case main body. In the end, generation of a crease in a separator or displacement of an electrode plate when the electrode body is accommodated can be suppressed.

[0008] The above and other elements, features, steps, characteristics and advantages of the present

disclosure will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a perspective view schematically illustrating an electrical energy storage device according to an embodiment;
[0010] FIG. **2** is a schematic longitudinal cross-sectional view taken along line II-II in FIG. **1**;
[0011] FIG. **3** is a schematic longitudinal cross-sectional view taken along line III-III in FIG. **1**;
[0012] FIG. **4** is a plan view of a plate material according to an embodiment;
[0013] FIG. **5A** is a side view schematically illustrating an aspect at pressing in a second surface preliminarily forming step;
[0014] FIG. **5B** is a side view schematically illustrating an aspect after the pressing in the second surface preliminarily forming step;
[0015] FIG. **6A** is a side view schematically illustrating an aspect when electrode bodies are disposed in a wrapping step;
[0016] FIG. **6B** is a side view schematically illustrating an aspect at pressing in the wrapping step;
[0017] FIG. **7** is a side view schematically illustrating an aspect in a bonding step;
[0018] FIG. **8** is a diagram corresponding to FIG. **3** when the number of electrode bodies is one according to a first modification;
[0019] FIG. **9** is a diagram corresponding to FIG. **3** when the number of electrode bodies is three according to a second modification; and
[0020] FIG. **10** is a diagram corresponding to FIG. **4** according to a third modification.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0021] Hereinafter, preferred embodiments of the art disclosed herein will be described with reference to the drawings as appropriate. Matters that are other than matters particularly mentioned in the present specification and that are necessary for the implementation of the present disclosure (for example, the general configuration and manufacturing process of an electrical energy storage device that do not characterize the present disclosure) can be grasped as design matters of those skilled in the art based on the prior art in the relevant field. The art disclosed herein can be implemented on the basis of the disclosure of the present specification and common technical knowledge in the relevant field. Note that in the drawings below, the members and parts with the same operation are denoted by the same reference signs and the overlapping description may be omitted or simplified. Moreover, in the present specification, the notation “A to B” for a range signifies a value more than or equal to A and less than or equal to B, and is meant to encompass also the meaning of being “more than A” and “less than B”.

Electrical Energy Storage Device

[0022] First, an electrical energy storage device manufactured by the art disclosed herein is described. Note that in the present specification, the term “electrical energy storage device” refers to general devices that are capable of being charged and discharged repeatedly, and corresponds to a concept encompassing secondary batteries such as lithium ion secondary batteries and nickel-hydrogen batteries and capacitors such as lithium ion capacitors and electrical double-layer capacitors.

[0023] FIG. **1** is a perspective view schematically illustrating an electrical energy storage device **100** according to an embodiment. FIG. **2** is a schematic longitudinal cross-sectional view taken along line II-II in FIG. **1**. FIG. **3** is a schematic longitudinal cross-sectional view taken along line III-III in FIG. **1**. As illustrated in FIG. **1**, the electrical energy storage device **100** has a square shape including a hexahedron (here, rectangular parallelepiped shape). In addition, in the description

below, reference signs F, Rr, L, R, U, and D in the drawings respectively denote front, rear, left, right, up, and down, and reference signs X, Y, and Z in the drawings respectively denote a thickness direction of the electrical energy storage device **100**, a width direction that is orthogonal to the thickness direction, and an up-down direction that is orthogonal to the thickness direction and the width direction. However, these are merely directions for convenience of description and do not limit the mode of installation of the electrical energy storage device **100**.

[0024] As illustrated in FIG. 2, the electrical energy storage device **100** includes a case **10**, one or a plurality of (two or more) electrode bodies **20**, a positive electrode terminal **30**, and a negative electrode terminal **40**. The electrical energy storage device **100** further includes a nonaqueous electrolyte solution (not illustrated) here. The electrical energy storage device **100** is a nonaqueous electrolyte solution secondary battery here, and is a lithium ion secondary battery, for example.

[0025] The case **10** is a housing that accommodates the electrode body **20**. Here, as illustrated in FIG. 1 and FIG. 2, the case **10** has an outer shape having a flat and bottomed square shape (rectangular parallelepiped shape here). The material of the case **10** may be the same as a material that has been used conventionally, and is not particularly limited. The case **10** is preferably made of metal and is more preferably formed of, for example, iron, an iron alloy such as stainless steel, aluminum, an aluminum alloy, or the like.

[0026] As illustrated in FIG. 2, here, the case **10** includes a case main body **12** with a square tubular shape having a pair of openings **12h** at both end parts in the width direction Y, and two sealing plates **14** that close the pair of openings **12h** of the case main body **12**. The case **10** is integrated in such a way that the sealing plates **14** are bonded (for example, bonded by welding) to peripheries of the pair of openings **12h** of the case main body **12**. The case **10** is hermetically sealed (closed).

[0027] As illustrated in FIG. 1, the case main body **12** includes a bottom surface **12a** with an approximately rectangular shape having long sides and short sides, a pair of long side surfaces **12b** extending from the pair of long sides (a pair of edges) of the bottom surface **12a** and facing each other, and a top surface **12c** with an approximately rectangular shape facing the bottom surface **12a**. The long side surface **12b** is larger in area than the bottom surface **12a** and the top surface **12c**. The top surface **12c** connects between upper end parts of the pair of long side surfaces **12b**. In this embodiment, the bottom surface **12a** is one example of “a first surface”, the pair of long side surfaces **12b** are one example of “a pair of second surfaces”, and the top surface **12c** is one example of “a third surface”. In another embodiment, however, the top surface **12c** may be the first surface and the bottom surface **12a** may be the third surface, for example.

[0028] Note that in the present specification, the term “substantially rectangular shape” encompasses, in addition to a perfect rectangular shape (rectangle), for example, a shape whose corner connecting a long side and a short side of the rectangular shape is rounded, a shape whose corner includes a notch, and the like.

[0029] The case main body **12** is formed in such a way that, here, one plate material **12p** (see FIG. 4) is bent into a square tubular shape and the joint is bonded (for example, bonded by welding), which will be described in detail below. In this embodiment, as illustrated in FIG. 1 and FIG. 3, a bonding part (preferably, welding bonding part) **12d** is provided at the top surface **12c** of the case main body **12** and this bonding part **12d** is formed by bonding the joint of the plate material **12p**. The bonding part **12d** is provided along a front corner part of the top surface **12c** (border with the long side surface **12b** on the front side).

[0030] The sealing plate **14** is a plate-shaped member that seals the opening **12h**. The sealing plate **14** has a substantially rectangular shape in a plan view. As illustrated in FIG. 1 and FIG. 2, the sealing plate **14** is provided with a liquid injection hole **15**. The liquid injection hole **15** is used to inject the electrolyte solution into the case **10** after the sealing plate **14** is assembled to the case main body **12**. The liquid injection hole **15** is sealed with a sealing member **16** after the electrolyte solution is injected. Note that although the liquid injection hole **15** is provided in the sealing plate

14 in this embodiment, the liquid injection hole **15** may be provided in the case main body **12** in another embodiment.

[0031] The electrode body **20** is accommodated inside the case **10** as illustrated in FIG. 2 and FIG. 3. The structure of the electrode body **20** may be similar to the conventional one without particular limitations. The electrode body **20** may be accommodated inside the case **10** while being covered with an insulating sheet (electrode body holder) made of resin. As illustrated in FIG. 3, here, two (a plurality of) electrode bodies **20** are accommodated inside one case **10**. The two electrode bodies **20** are disposed along the thickness direction (arrangement direction) X. However, the number of electrode bodies **20** to be accommodated inside one case **10** is not limited in particular and may be one, or three or more in another embodiment.

[0032] The electrode body **20** is here a wound electrode body in which a positive electrode with a band shape and a negative electrode with a band shape are stacked through a separator with a band shape and wound in a longitudinal direction using a winding axis as a center. However, in another embodiment, the electrode body **20** may be a multilayer electrode body in which a plurality of positive electrodes with a square shape (typically, a rectangular shape) and a plurality of negative electrodes with a square shape (typically, a rectangular shape) are stacked on each other in an insulated state.

[0033] As illustrated in FIG. 3, the electrode body **20** has an outer shape that is flat here. The electrode body **20** includes a pair of curved parts **20r** and a pair of flat parts **20f** coupling the pair of curved parts **20r**. In this embodiment, the electrode body **20** is disposed inside the case **10** with the winding axis substantially parallel to the width direction Y. Therefore, the pair of curved parts **20r** of the electrode body **20** face the bottom surface **12a** and the top surface **12c** of the case main body **12**. The pair of flat parts **20f** of the electrode body **20** face the pair of long side surfaces **12b** of the case main body **12**. However, in another embodiment, the electrode body **20** may be disposed inside the case **10** with the winding axis substantially parallel to the up-down direction Z, for example.

[0034] As illustrated in FIG. 2, a positive electrode tab **23** with a convex shape is provided in the positive electrode of the electrode body **20**. The positive electrode tab **23** is electrically connected to the positive electrode terminal **30** through a positive electrode current collecting part **32**. A negative electrode tab **24** with a convex shape is provided in the negative electrode of the electrode body **20**. The negative electrode tab **24** is electrically connected to the negative electrode terminal **40** through a negative electrode current collecting part **42**.

[0035] The nonaqueous electrolyte solution is accommodated inside the case **10** together with the electrode body **20**. The nonaqueous electrolyte solution may be similar to the conventional one without particular limitations. The nonaqueous electrolyte solution typically contains a nonaqueous solvent and a supporting salt (electrolyte salt, for example Li salt or Na salt). The nonaqueous electrolyte solution is typically in a liquid form but may alternatively be in a gel form. In another embodiment, the electrical energy storage device **100** may contain a solid electrolyte instead of the nonaqueous electrolyte solution. In this case, the separator can be omitted.

[0036] The positive electrode terminal **30** and the negative electrode terminal **40** are fixed to the respective surfaces facing the case **10** (specifically, the pair of sealing plates **14**) here. The positive electrode terminal **30** is attached to the sealing plate **14** disposed on one side in the width direction Y (on the right side in FIG. 1 and FIG. 2). The negative electrode terminal **40** is attached to the sealing plate **14** disposed on the other side in the width direction Y (on the left side in FIG. 1 and FIG. 2). In this embodiment, the positive electrode terminal **30** and the negative electrode terminal **40** are provided at the sealing plates **14**; however, the positive electrode terminal **30** and the negative electrode terminal **40** may alternatively be provided at the case main body **12** in another embodiment.

[0037] The positive electrode terminal **30** is preferably made of metal, and is more preferably formed of aluminum or an aluminum alloy, for example. The negative electrode terminal **40** is

preferably made of metal and is more preferably formed of copper or a copper alloy, for example. As illustrated in FIG. 2, the positive electrode terminal **30** is electrically connected to the positive electrode tab **23** of the electrode body **20** through the positive electrode current collecting part **32** inside the case **10**. The negative electrode terminal **40** is electrically connected to the negative electrode tab **24** of the electrode body **20** through the negative electrode current collecting part **42** inside the case **10**.

Manufacturing Method for Electrical Energy Storage Device

[0038] The electrical energy storage device **100** can be manufactured suitably by a manufacturing method including, for example, a preparing step (step S1), a second surface preliminarily forming step (S2), a wrapping step (step S3), a plate material bonding step (step S4), and a sealing plate bonding step (step S5) typically in this order. However, the second surface preliminarily forming step (S2) is not essential and can be omitted in another embodiment. The manufacturing method according to this embodiment may include other step than those described above at an optional stage as appropriate. Each step will be described below.

[0039] The preparing step (step S1) is a step of preparing a plate material made of metal that constitutes at least a part of the case main body **12**. FIG. 4 is a plan view of the plate material **12p** according to one embodiment. The plate material **12p** is preferably formed of, for example, iron, an iron alloy such as stainless steel, aluminum, an aluminum alloy, or the like. The plate material **12p** has a flat plate shape here and has the thickness that is substantially uniform. The thickness of the plate material **12p** is preferably about 0.1 to 2 mm, more preferably 0.2 to 1 mm, and still more preferably 0.4 to 0.8 mm from the viewpoints of mechanical strength, rigidity, durability, and the like, although there is no particular limitation. The thickness of the plate material **12p** may be the same as the thickness of the sealing plate **14**, for example, or may be larger than the thickness of the sealing plate **14**.

[0040] As illustrated in FIG. 4, the plate material **12p** has an approximately rectangular shape here. The plate material **12p** includes at least a first part **P1** that forms the bottom surface **12a** (first surface) of the case **10**, and a pair of second parts **P2** that extend from the first part **P1** and form the pair of long side surfaces **12b** (the pair of second surfaces) of the case **10**. The first part **P1** has the same shape and size as those of the bottom surface **12a**. The first part **P1** has an approximately rectangular shape having long sides and short sides here. The second part **P2** has the same shape and size as those of the long side surface **12b**. The second parts **P2** here extend from the pair of long sides of the first part **P1**. Note that in FIG. 4, reference signs L_x , L_y , and L_z denote the lengths respectively coinciding with the length (thickness) of the outer shape of the electrical energy storage device **100** in the thickness direction X, the length (width) thereof in the width direction Y, and the length (height) thereof in the up-down direction Z.

[0041] The plate material **12p** preferably further includes one or two third parts **P3** that extend from at least one of the pair of second parts **P2** and form the top surface **12c** (third surface) of the case **10**. In FIG. 4, the plate material **12p** further includes one third part **P3** that extends from one second part **P2** and forms the top surface **12c** of the case **10**. The third part **P3** has the same shape and size as those of the top surface **12c** here. The third part **P3** extends from the long side of one second part **P2** here. The first part **P1**, the second part **P2**, and the third part **P3** have the long sides with the same length L_y .

[0042] In FIG. 4, border lines between the respective parts are expressed by dashed lines and these border lines virtually indicate the parts to be bent in the second surface preliminarily forming step (step S2) and/or the wrapping step (step S3) to be described below. In another embodiment, however, a notch, a guide, or the like may be provided at the border line, for example, so as to make it easy to bend the plate material **12p** in the second surface preliminarily forming step (step S2) and/or the wrapping step (step S3).

[0043] The second surface preliminarily forming step (step S2) is a step of bending the border between the first part **P1** and the second part **P2** of the plate material **12p** at a bending angle larger

than 90°. In this embodiment, the work from this step to the plate material bonding step (step S4) is performed using a manufacturing apparatus **200** (see FIG. 5A, etc.). In another embodiment, however, a part of or all the steps may be performed using another apparatus or performed manually. FIG. 5A is a side view schematically illustrating an aspect at pressing in this step. FIG. 5B is a side view schematically illustrating an aspect after the pressing in this step.

[0044] In this step, first, the plate material **12p** prepared in the preparing step (step S1) is disposed on a formation stage **210** of the manufacturing apparatus **200** (see FIG. 5A, etc.). Next, as illustrated in FIG. 5A, after a pressing plate **120** is disposed at the first part **P1** of the plate material **12p**, the pair of second parts **P2** are held between a pair of side surface pressurizing plates **130**. The side surface pressurizing plates **130** are electrically connected to a control unit that is not illustrated here, and by the control unit, the side surface pressurizing plates **130** can be transferred between a pressed state in FIG. 5A (the pair of second parts **P2** are pressed) and a retracted state in FIG. 5B (the pair of second parts **P2** are not pressed). Next, as indicated by arrows in FIG. 5A, the pair of side surface pressurizing plates **130** are pressed to apply a predetermined amount of pressure on the plate material **12p**; thus, the borders between the first part **P1** and the second parts **P2** are bent so that the pair of second parts **P2** get close to each other. Accordingly, the bottom surface **12a** (first surface) of the case **10** including the first part **P1** is defined.

[0045] After the pressurization is completed, the pair of side surface pressurizing plates **130** are retracted and the pressing plate **120** is extracted as indicated by arrows in FIG. 5B. In this embodiment, the plate material **12p** is bent so that an angle (bending angle) $\theta 1$ between the first part **P1** and the second part **P2** becomes larger than 90°. Thus, the interference between the electrode body **20** and the second part **P2** of the plate material **12p** in the subsequent wrapping step (step S3) can be prevented and the accommodation property of the electrode body **20** can be improved more.

[0046] The wrapping step (step S3) is a step of wrapping the electrode body **20** by disposing one or the plurality of electrode bodies **20** on the first part **P1** of the plate material **12p** and bending the plate material **12p** so that the distance between the pair of second parts **P2** becomes narrow. FIG. 6A is a side view schematically illustrating an aspect when the electrode bodies are disposed in this step. FIG. 6B is a side view schematically illustrating an aspect at the pressing in this step.

[0047] In this step, the electrode body **20** is prepared first. In this embodiment, the sealing plate **14** is attached to the electrode body **20** before the plate material bonding step (step S4), which will be described below, so that an integrated object of the electrode body **20** and the sealing plate **14** is manufactured. Specifically, the positive electrode current collecting part **32** attached to the positive electrode tab **23** of the electrode body **20** is bonded to the positive electrode terminal **30** fixed to the sealing plate **14**, and the negative electrode current collecting part **42** attached to the negative electrode tab **24** of the electrode body **20** is bonded to the negative electrode terminal **40** fixed to the sealing plate **14**. The bonding can be performed by welding such as laser welding, ultrasonic welding, or resistance welding, for example. Note that the electrode body **20** may be covered with an insulating sheet. In another embodiment, the sealing plate **14** may be attached after the plate material bonding step (step S4), for example. In one example, the sealing plate **14** may be attached in the sealing plate bonding step (step S5).

[0048] Next, as indicated by an arrow in FIG. 6A, the electrode bodies **20** are disposed on the first part **P1** of the plate material **12p**. In this embodiment, the integrated object of the electrode bodies **20** and the sealing plate **14** (the illustration is omitted in FIG. 6A) is disposed on the first part **P1** of the plate material **12p**. The electrode bodies **20** (or the integrated object) are entirely accommodated below an upper end of the second part **P2** of the plate material **12p**. As described above, in this embodiment, the plurality of (specifically, two) electrode bodies **20** are used. Each of the plurality of electrode bodies **20** is a flat wound electrode body including the pair of curved parts **20r** and the pair of flat parts **20f** coupling the pair of curved parts **20r**. The plurality of electrode bodies **20** are disposed on the first part **P1** from above the second parts **P2** of the plate material **12p**

here. Each of the plurality of electrode bodies **20** is disposed along the thickness direction (arrangement direction) **X** so that one curved part **20r** (the lower one in FIG. 6A) faces the first part **P1** of the plate material **12p**.

[0049] Next, as indicated by arrows in FIG. 6B, the pair of side surface pressurizing plates **130** are pressed again to apply a predetermined amount of pressure on the plate material **12p** so that the distance between the pair of second parts **P2** is shortened; thus, the borders between the first part **P1** and the second parts **P2** are bent. Accordingly, an angle (bending angle) θ_2 between the first part **P1** and the second part **P2** becomes approximately 90° , and the pair of flat parts **20f** of the electrode body **20** become substantially parallel to the second parts **P2** of the plate material **12p**. In this manner, the pair of long side surfaces **12b** (pair of second surfaces) of the case **10** formed of the pair of second parts **P2** are defined. One curved part **20r** (lower one in FIG. 6A) and the pair of flat parts **20f** of the electrode body **20** are wrapped with the first part **P1** and the pair of second parts **P2** of the plate material **12p** from three directions.

[0050] The plate material bonding step (step S4) is a step of bending the third part **P3** of the plate material **12p** so as to face the first part **P1**, thereby forming the plate material **12p** into a square tubular shape and bonding the joint in a linear shape. In the plate material bonding step (step S4) in this embodiment, after the plate material **12p** is formed into the square tubular shape, an end part **E3** (see FIG. 4 and FIG. 6B) of the third part **P3** of the plate material **12p** and an end part **E2** (see FIG. 4 and FIG. 6B) of the second part **P2** that is in contact with the end part **E3** are bonded to each other. FIG. 7 is a side view schematically illustrating an aspect in this step. In this embodiment, the pair of side surface pressurizing plates **130** keep pressing the pair of second parts **P2** (pressed state) continuously from the wrapping step (step S3).

[0051] In this step, as illustrated in FIG. 7, the third part **P3** of the plate material **12p** is pressed with a top surface pressurizing plate **140** first and a predetermined amount of pressure is applied to the top surface pressurizing plate **140** as indicated by an arrow in FIG. 7, so that a border between the second part **P2** and the third part **P3** is bent. Thus, the angle (bending angle) between the second part **P2** and the third part **P3** becomes approximately 90° and the third part **P3** becomes substantially parallel to the first part **P1**. In this manner, the top surface **12c** (third surface) of the case **10** that is formed of the third part **P3** is defined.

[0052] Next, as indicated by a circular mark in FIG. 7, the joint of the plate material **12p** (specifically, the end part **E3** of the third part **P3** and the end part **E2** of the second part **P2** illustrated in FIG. 4 and FIG. 6B) is bonded in a linear shape. Note that the bonding method for the plate material **12p** is not limited in particular and may be similar to the conventional method. The bonding can be performed by welding such as laser welding, ultrasonic welding, or resistance welding, for example. In addition, the bonding conditions may be similar to the conventional ones without particular limitations.

[0053] In some embodiments, the bonding is preferably performed with the pair of second parts **P2** of the plate material **12p** pressed from both sides. In this embodiment, the bonding is performed in a state where the pair of second parts **P2** of the plate material **12p** are sandwiched between the pair of side surface pressurizing plates **130** and pressed thereby from both sides. Thus, it is possible to suppress that the bending angle between the first part **P1** and the second part **P2** becomes more than 90° , and at the time of bonding, the substantially parallel state between the pair of second parts **P2** can be kept easily. Accordingly, the formability and the size accuracy can be improved.

[0054] In this embodiment, the bonding is performed further with the third part **P3** of the plate material **12p** pressed by the top surface pressurizing plate **140**. Thus, it is possible to suppress that the bending angle between the second part **P2** and the third part **P3** becomes more than 90° , and at the time of bonding, the substantially parallel state between the third part **P3** and the first part **P1** can be kept easily. Accordingly, the formability and the size accuracy can be improved.

[0055] When the electrode body **20** is a wound electrode body with a flat shape, the bonding is preferably performed at a position that is a corner part of the case **10** or at a position that is L/N

from the corner part (in which N represents the number of electrode bodies **20** disposed in the case **10** and is an integer of 2 or more, and L represents the length of the case **10** in the thickness direction (arrangement direction) X of the electrode body **20**). When the number of electrode bodies **20** disposed in one case **10** is 2 (N=2) as described in this embodiment, for example, the bonding is preferably performed at the position that is the corner part of the case **10**, or at a position that is L/2 from the corner part in the thickness direction X (that is, at a center in the thickness direction X). Thus, the heat at the bonding conducts less easily to an upper end part of the electrode body **20**, specifically to the curved part **20r** on the side closer to the third part P3 (upper side in FIG. 7) and the thermal damage on the electrode body **20** can be suppressed.

[0056] In this embodiment, as illustrated in FIG. 7, the welding is performed at the corner part of the case **10**, specifically along the front corner part of the top surface **12c** (the border between the second part P2 and the third part P3). Thus, the linear bonding part **12d** is provided as illustrated in FIG. 1 and FIG. 3. In this manner, the case main body **12** with the square tubular shape including the plate material **12p** can be obtained.

[0057] In the sealing plate bonding step (step S5), two sealing plates **14** are prepared and bonded (for example, bonded by welding) to the periphery of the pair of openings **12h** of the case main body **12** (see FIG. 2). The sealing plate **14** may be similar to the conventional one without particular limitations. The sealing plate **14** is preferably made of metal, and is preferably made of the same kind of metal as the plate material **12p**, for example. The bonding method for the sealing plate **14** is not limited in particular and may be similar to the conventional method. The bonding of the sealing plate **14** can be performed by welding such as laser welding, ultrasonic welding, or resistance welding, for example. After the opening **12h** is sealed with the sealing plate **14**, the nonaqueous electrolyte solution is injected typically through the liquid injection hole **15** provided in the sealing plate **14** or the like. Then, the case **10** is sealed by closing the liquid injection hole **15** with the sealing member **16**. In this manner, the electrical energy storage device **100** can be manufactured.

Application of Electrical Energy Storage Device

[0058] The electrical energy storage device **100** can be used in various applications, and for example, suitably used as a motive power source (electrical power source for driving) for a motor mounted on a vehicle such as a passenger car or a truck. Although the type of vehicles is not particularly limited, examples thereof may include a plug-in hybrid electric vehicle (PHEV), a hybrid electric vehicle (HEV), a battery electric vehicle (BEV), and the like.

[0059] Although the preferable embodiments of the present disclosure have been described above, they are merely examples. The present disclosure can be implemented in various other modes. The present disclosure can be implemented based on the contents disclosed in the present specification and the technical common sense in the relevant field. The techniques described in the scope of claims include those in which the embodiments exemplified above are variously modified and changed. For example, another modification can replace a part of the aforementioned embodiment or be added to the aforementioned embodiment. Additionally, the technical feature may be deleted as appropriate unless such a feature is described as an essential element.

[0060] For example, the aforementioned manufacturing method may further include a third surface preliminarily forming step of bending the border between the second part P2 and the third part P3 of the plate material **12p** after the preparing step (step S1) and before the wrapping step (step S3). In this case, it is preferable to exclude the second surface preliminarily forming step (step S2) from the viewpoint of preventing the interference between the bent third part P3 and the electrode body **20** in the wrapping step (step S3).

[0061] For example, the aforementioned manufacturing method includes the second surface preliminarily forming step (step S2) before the wrapping step (step S3), and in the second surface preliminarily forming step (step S2), the border between the first part P1 and the second part P2 is bent to make a crease in advance. However, the present disclosure is not limited to this example. In

a case where the border between the first part P1 and the second part P2 has a notch, a guide, or the like provided in advance, for example, the second surface preliminarily forming step (step S2) may be omitted and the plate material with the flat plate shape may be subjected to the wrapping step (step S3) directly.

[0062] For example, in the embodiment described above, the electrode body 20 is a wound electrode body with the flat shape, and in the plate material bonding step (step S4), the bonding is performed at the position that is the corner part of the case 10 or at the position that is L/N from the corner part. However, the present disclosure is not limited to this example. In a modification, the electrode body 20 may be a multilayer electrode body. In this case, the bonding is preferably performed at the position that is the corner part of the case 10 in the plate material bonding step (step S4). Thus, the heat at the bonding conducts less easily to the upper end part of the electrode body 20, and the thermal damage on the electrode body 20 can be suppressed.

[0063] For example, in the embodiment described above, the two electrode bodies 20 are used as illustrated in FIG. 3, etc. However, the present disclosure is not limited to this example. The number of electrode bodies 20 to be accommodated in one case 10 may be one, or three or more.

[0064] FIG. 8 is a diagram corresponding to FIG. 3 according to a first modification in which the number of electrode bodies 20 is one. Note that the portions corresponding to those in the plate material 12p are denoted in parenthesis here. In a case where the number of electrode bodies 20 to be disposed in one case 10 is one, the bonding in the plate material bonding step (step S4) is preferably performed at the position that is the corner part of the case 10, here, at the position that is a front corner part A1 or a rear corner part A2 of the top surface 12c of the case 10 as indicated by circles in FIG. 8. Although the bonding is performed at the position that is the top surface 12c here, the bonding may alternatively be performed at the position that is the bottom surface 12a, for example.

[0065] FIG. 9 is a diagram corresponding to FIG. 3 according to a second modification in which the number of electrode bodies 20 is three. In a case where the number of electrode bodies 20 to be disposed in one case 10 is three (N=3), the bonding in the plate material bonding step (step S4) is preferably performed at the position that is the corner part of the case 10 or at a position that is L/3 from the corner part in the thickness direction X. Here, it is preferable to perform the bonding at any one of positions among a position that is a front corner part B1, a position that is a rear corner part B2, and positions B3 and B4 that are L/3 from the corner parts B1 and B2 in the thickness direction X on the top surface 12c of the case 10 as indicated by circles in FIG. 9. Although the bonding is performed at the position that is the top surface 12c here, the bonding may alternatively be performed at the position that is the bottom surface 12a, for example.

[0066] In the plate material 12p in FIG. 4 described above, for example, one third part P3 is used and has the same shape and size as those of the top surface 12c. However, the present disclosure is not limited to this example. FIG. 10 is a diagram corresponding to FIG. 4 according to a third modification. As illustrated in FIG. 10, a plate material 12p1 includes two third parts P31 and P32. The two third parts P3 extend from the long sides of the pair of second parts P2 here. Bonding the two third parts P31 and P32 along the joint makes the same shape and size as those of the top surface 12c.

[0067] For example, when the number of electrode bodies 20 to be disposed in one case 10 is two (N=2), the length of each of the two third parts P31 and P32 in the short side direction is Lx/2. In this modification, in the plate material bonding step (step S4), an end part E31 of the third part P31 and an end part E32 of the third part P32 in the plate material 12p1 are bonded together. Thus, the bonding part is provided in a linear shape at a center of the top surface 12c of the case main body 12 in the thickness direction X, for example.

[0068] As described above, the following items are given as specific aspects of the art disclosed herein.

[0069] Item 1: The manufacturing method for an electrical energy storage device including one or

the plurality of electrode bodies and the case that has the square shape and accommodates the electrode body or the electrode bodies, in which the case includes at least the first surface, the pair of second surfaces extending from the pair of edges of the first surface and facing each other, and the third surface facing the first surface, the manufacturing method including: the preparing step of preparing the plate material that is made of metal and includes at least the first part forming the first surface of the case, and the pair of second parts extending from the first part and forming the pair of second surfaces of the case; and the wrapping step of disposing the electrode body on the first part of the plate material and bending the plate material so that the distance between the pair of second parts decreases, thereby wrapping the electrode body.

[0070] Item 2: The manufacturing method for an electrical energy storage device according to Item 1, in which the plate material prepared in the preparing step further includes one or two third parts that extend from at least one of the pair of second parts and form the third surface of the case.

[0071] Item 3: The manufacturing method for an electrical energy storage device according to Item 2, further including, after the wrapping step, the plate material bonding step of bending the third part of the plate material so as to face the first part and to form the plate material into the square tubular shape, and bonding the joint in the linear shape.

[0072] Item 4: The manufacturing method for an electrical energy storage device according to Item 3, in which in the plate material bonding step, the bonding is performed with the pair of second parts of the plate material pressed from both sides.

[0073] Item 5: The manufacturing method for an electrical energy storage device according to Item 3 or 4, in which the plurality of electrode bodies are used, and each of the plurality of electrode bodies is the wound electrode body with the flat shape including the pair of curved parts and the pair of flat parts coupling the pair of curved parts, in the wrapping step, each of the plurality of wound electrode bodies is disposed along the arrangement direction so that one of the curved parts faces the first part, and in the plate material bonding step, the bonding is performed at the position that is the corner part of the case or at the position that is L/N from the corner part (in which N represents the number of electrode bodies disposed in the case and is an integer of 2 or more, and L represents the length of the case in the arrangement direction of the electrode bodies).

[0074] Item 6: The manufacturing method for an electrical energy storage device according to Item 3 or 4, in which the electrode body is the multilayer electrode body, and in the plate material bonding step, the bonding is performed at the position that is the corner part of the case.

[0075] Item 7: The manufacturing method for an electrical energy storage device according to any one of Items 1 to 6, further including, after the preparing step and before the wrapping step, the second surface preliminarily forming step of bending the border between the first part and the second part of the plate material with a bending angle of more than 90°.

[0076] Item 8: The manufacturing method for an electrical energy storage device according to any one of Items 1 to 6, further including, after the preparing step and before the wrapping step, the third surface preliminarily forming step of bending the border between the second part and the third part of the plate material.

REFERENCE SIGNS LIST

[0077] **10** Case [0078] **12** Case main body [0079] **12a** Bottom surface (first surface) [0080] **12b** Long side surface (second surface) [0081] **12c** Top surface (third surface) [0082] **12d** Bonding part [0083] **14** Sealing plate [0084] **20** Electrode body [0085] **20f** Flat part [0086] **20r** Curved part [0087] **100** Electrical energy storage device [0088] **12p**, **12p1** Plate material [0089] **P1** First part [0090] **P2** Second part [0091] **P3**, **P31**, **P32** Third part

Claims

1. A manufacturing method for an electrical energy storage device including one or a plurality of electrode bodies and a case that has a square shape and accommodates the electrode body or the

electrode bodies, in which the case includes at least a first surface, a pair of second surfaces extending from a pair of edges of the first surface and facing each other, and a third surface facing the first surface, the manufacturing method comprising: a preparing step of preparing a plate material that is made of metal and includes at least a first part forming the first surface of the case, and a pair of second parts extending from the first part and forming the pair of second surfaces of the case; and a wrapping step of disposing the electrode body on the first part of the plate material and bending the plate material so that a distance between the pair of second parts decreases, thereby wrapping the electrode body.

2. The manufacturing method for an electrical energy storage device according to claim 1, wherein the plate material prepared in the preparing step further includes one or two third parts that extend from at least one of the pair of second parts and form the third surface of the case.

3. The manufacturing method for an electrical energy storage device according to claim 2, further comprising, after the wrapping step, a plate material bonding step of bending the third part of the plate material so as to face the first part and to form the plate material into a square tubular shape, and bonding a joint in a linear shape.

4. The manufacturing method for an electrical energy storage device according to claim 3, wherein in the plate material bonding step, the bonding is performed with the pair of second parts of the plate material pressed from both sides.

5. The manufacturing method for an electrical energy storage device according to claim 3, wherein the plurality of electrode bodies are used, and each of the plurality of electrode bodies is a wound electrode body with a flat shape including a pair of curved parts and a pair of flat parts coupling the pair of curved parts, in the wrapping step, each of the plurality of wound electrode bodies is disposed along an arrangement direction so that one of the curved parts faces the first part, and in the plate material bonding step, the bonding is performed at a position that is a corner part of the case or at a position that is L/N from the corner part (in which N represents the number of electrode bodies disposed in the case and is an integer of 2 or more, and L represents a length of the case in the arrangement direction of the electrode bodies).

6. The manufacturing method for an electrical energy storage device according to claim 3, wherein the electrode body is a multilayer electrode body, and in the plate material bonding step, the bonding is performed at a position that is a corner part of the case.

7. The manufacturing method for an electrical energy storage device according to claim 1, further comprising, after the preparing step and before the wrapping step, a second surface preliminarily forming step of bending a border between the first part and the second part of the plate material with a bending angle of more than 90° .

8. The manufacturing method for an electrical energy storage device according to claim 2, further comprising, after the preparing step and before the wrapping step, a third surface preliminarily forming step of bending a border between the second part and the third part of the plate material.
